Functional Specification Document

Sistem FingerScan Biometrik

Dokumen	Nilai
Nomor Dokumen	FSD-BJBS-FingerScan-01
Versi	1.1.0
Tanggal	30 Januari 2026
Referensi PRD	PRD-BJBS-FingerScan-01

Daftar Isi

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1. Pendahuluan

1.1 Tujuan Dokumen

Dokumen ini menjelaskan spesifikasi fungsional lengkap untuk **Sistem FingerScan Biometrik** yang akan digunakan sebagai mekanisme otorisasi transaksi di seluruh jaringan kantor Bank. Dokumen ini menjadi acuan bagi tim pengembang, QA, dan stakeholder terkait dalam proses development dan testing.

1.2 Ruang Lingkup

Dalam Scope:

No	Fitur	Keterangan
1	Enrollment biometrik	Pendaftaran sidik jari user definitif dan alternate
2	Verifikasi biometrik	Proses verifikasi untuk approval transaksi
3	Device management	Pengelolaan perangkat DigitalPersona 4500
4	Fallback mechanism	Mekanisme fallback ke password di modul DAF/BDS
5	Audit trail	Pencatatan seluruh aktivitas verifikasi
6	Integration API	Service untuk integrasi dengan surrounding apps

Di Luar Scope:

- Biometrik selain sidik jari (face recognition, iris scan)
- Login sistem menggunakan fingerprint
- Proses integrasi detail di sisi surrounding apps
- Fingerscan Management / Service tidak menangani proses fallback password

1.3 Definisi dan Istilah

Istilah	Definisi
Enrollment	Proses pendaftaran sidik jari user ke dalam sistem
Template	Data digital hasil ekstraksi fitur sidik jari (bukan gambar mentah)
Fingerstation	Aplikasi client yang terhubung dengan device fingerprint
Maker	User yang melakukan input transaksi
Checker/Approver	User yang melakukan otorisasi transaksi
Definitif	User utama yang memiliki wewenang approval
Alternate	User pengganti yang dapat melakukan approval
Fallback	Mekanisme cadangan menggunakan password
mTLS	Mutual TLS untuk autentikasi dua arah

1.4 Referensi

No	Dokumen	Versi
1	PRD-BJBS-FingerScan-01	1.0.0
2	DigitalPersona 4500 SDK Documentation	-
3	Bank Security Policy	-

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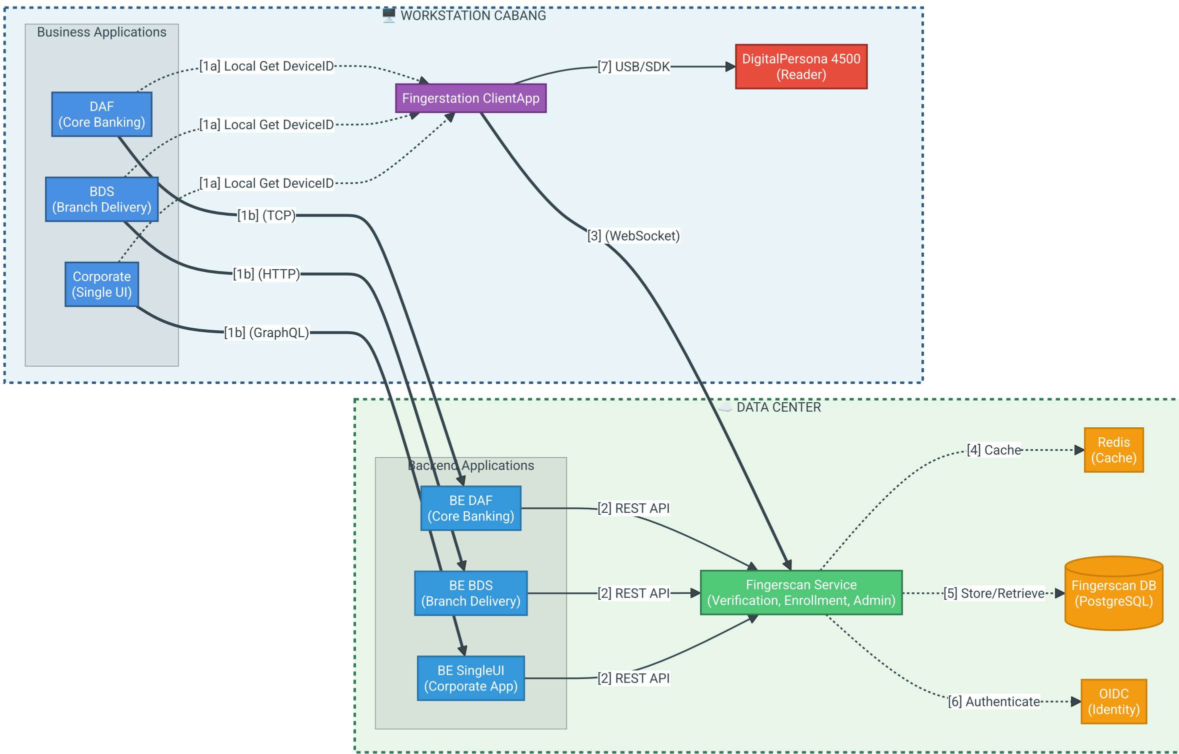
2. Deskripsi Umum Sistem

Sistem FingerScan Biometrik merupakan sistem middleware yang berfungsi sebagai layer verifikasi biometrik antara aplikasi bisnis (DAF, BDS, dll) dengan proses approval transaksi. Sistem ini menggantikan mekanisme otorisasi berbasis password untuk meningkatkan keamanan dan memastikan kehadiran fisik approver.

2.1 High Level Architecture



Penjelasan Alur:

No	Layer	Lokasi	Protokol	Deskripsi
[1a]	Apps → Fingerstation	**Local** (Workstation)	IPC/Local API	Business apps memanggil Fingerstation secara lokal untuk mendapatkan Device ID
[1b]	Apps → Backend Apps	**Network** (WAN/VPN)	TCP/HTTP/GraphQL	Business apps mengirim request otorisasi/enrollment

No	Layer	Lokasi	Protokol	Deskripsi
				ke Backend masing-masing
[2]	Backend → Service	**Network** (WAN/VPN)	REST API (HTTPS)	Backend apps mengirim request verifikasi ke Fingerscan Service
[3]	Fingerstation → Service	**Network** (WAN/VPN)	WebSocket (mTLS)	Fingerstation mengirim template hasil capture ke service via koneksi aman
[4]	Service → Cache	**Local** (Data Center)	Redis Protocol	Service mengecek cache untuk template yang sering digunakan
[5]	Service → Database	**Local** (Data Center)	JDBC/SQL	Service menyimpan dan mengambil data template, audit log, dan konfigurasi
[6]	Service → OIDC	**Local** (Data Center)	OAuth2/OIDC	Service melakukan autentikasi dan autorisasi user saat enrollment
[7]	Fingerstation → Device	**Local** (Workstation)	USB/SDK	Fingerstation berkomunikasi dengan hardware fingerprint reader

Catatan Arsitektur:

- 🖥️ **Workstation/Cabang**: Business applications (DAF, BDS, CMS) dan Fingerstation berjalan di PC yang sama
- ☁️ **Data Center/Kantor Pusat**: Fingerscan Service dan semua komponen backend berjalan secara terpusat
- 🔗 **Koneksi Lokal** ([1a], [6], [7]): Komunikasi dalam satu mesin/workstation
- 🌐 **Koneksi Network** ([1b], [2]): Komunikasi melalui jaringan (VPN/MPLS) antara cabang dan kantor pusat

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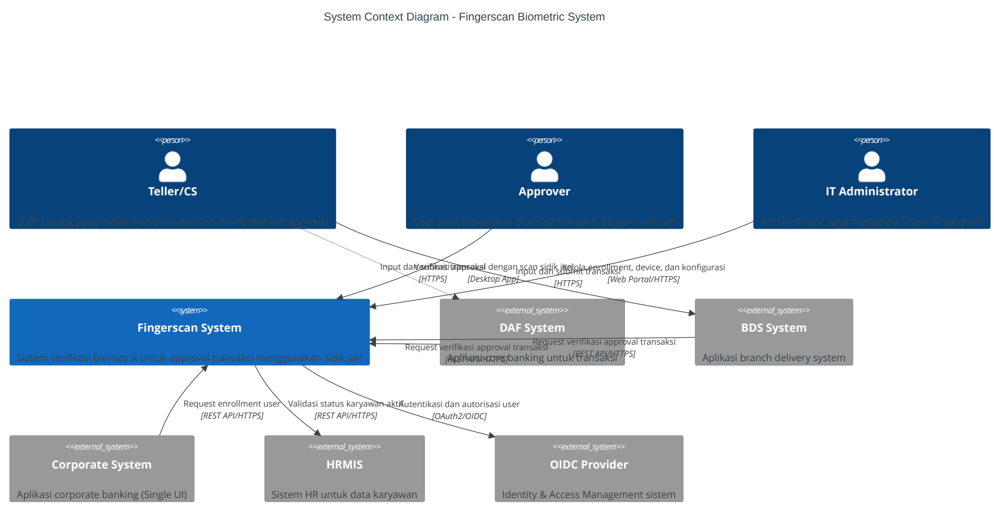
2.1.1 Context Diagram



Penjelasan Context Diagram:

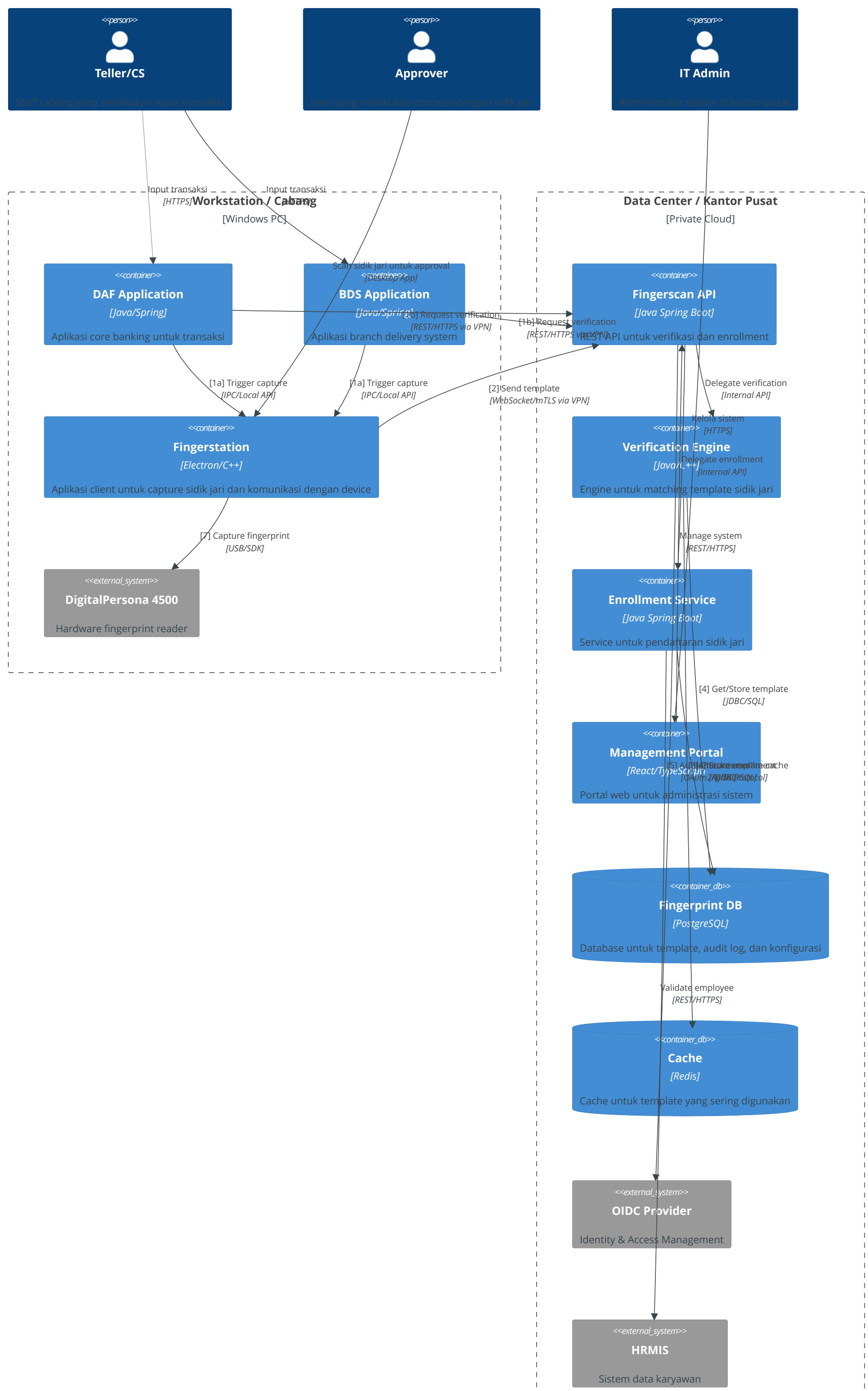
Elemen	Type	Deskripsi
Teller/CS	Person	User yang melakukan input transaksi dan meminta approval
Approver	Person	User yang melakukan otorisasi transaksi menggunakan sidik jari
IT Administrator	Person	Administrator yang mengelola enrollment, device, dan sistem

Elemen	Type	Deskripsi
Fingerscan System	System	Sistem inti untuk verifikasi biometrik dan manajemen enrollment
DAF System	External System	Core banking application yang memerlukan approval biometrik
BDS System	External System	Branch delivery system yang memerlukan approval biometrik
Corporate System	External System	Corporate banking app (Single UI) untuk enrollment
HRMIS	External System	Sistem HR untuk validasi data karyawan
OIDC Provider	External System	Identity provider untuk autentikasi dan autorisasi

Key Interactions: - Business Apps (DAF, BDS) mengirim request verifikasi ke Fingerscan System - Approver berinteraksi langsung dengan Fingerscan untuk scan sidik jari - Fingerscan System memvalidasi user melalui HRMIS dan OIDC - Administrator mengelola sistem melalui web portal

2.1.2 Container Diagram

Container Diagram - Fingerscan Biometric System



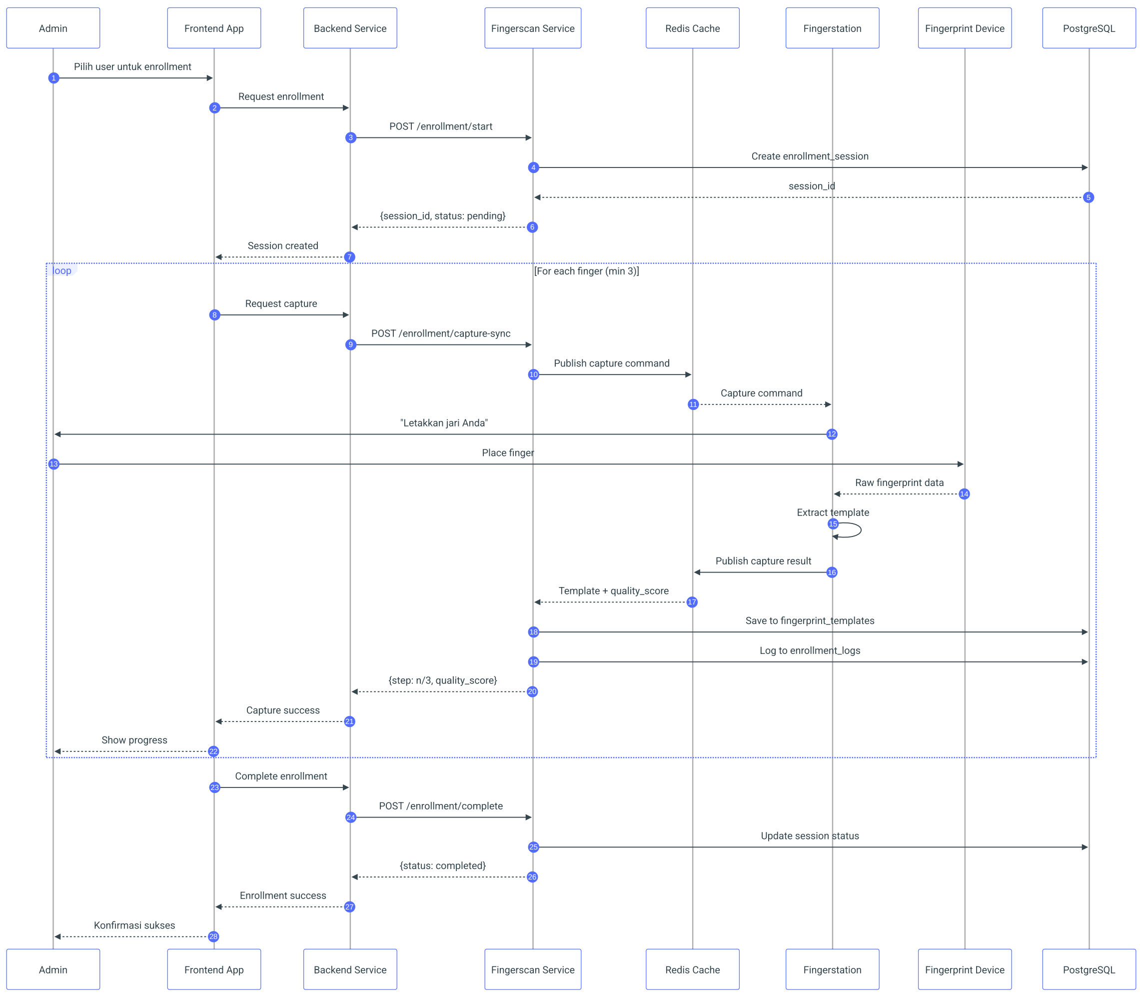
Penjelasan Container Diagram:

Elemen	Jenis	Deskripsi
Teller/CS	Person	User yang melakukan input transaksi di aplikasi bisnis
Approver	Person	User yang melakukan otorisasi menggunakan sidik jari
IT Admin	Person	Administrator yang mengelola sistem fingerprint
Workstation Boundary	Boundary	PC di cabang yang menjalankan business apps dan fingerstation
Data Center Boundary	Boundary	Infrastruktur terpusat di kantor pusat
DAF/BDS	Container	Aplikasi bisnis yang memerlukan otorisasi fingerprint
Fingerstation	Container	Aplikasi client untuk interface dengan device
DigitalPersona 4500	External System	Hardware fingerprint reader
Fingerscan API	Container	API gateway untuk semua request verifikasi/enrollment
Verification Engine	Container	Engine untuk matching sidik jari (1:1 matching)
Enrollment Service	Container	Service untuk proses pendaftaran sidik jari
Management Portal	Container	Web portal untuk administrasi

Elemen	Jenis	Deskripsi
Fingerprint DB	Database	PostgreSQL untuk menyimpan template dan audit
Cache	Database	Redis untuk performa (caching template)
OIDC Provider	External System	SSO untuk autentikasi user
HRMIS	External System	Sistem HR untuk validasi karyawan

2.2 High Level Process Flow

2.2.1 Enrollment Process Flow



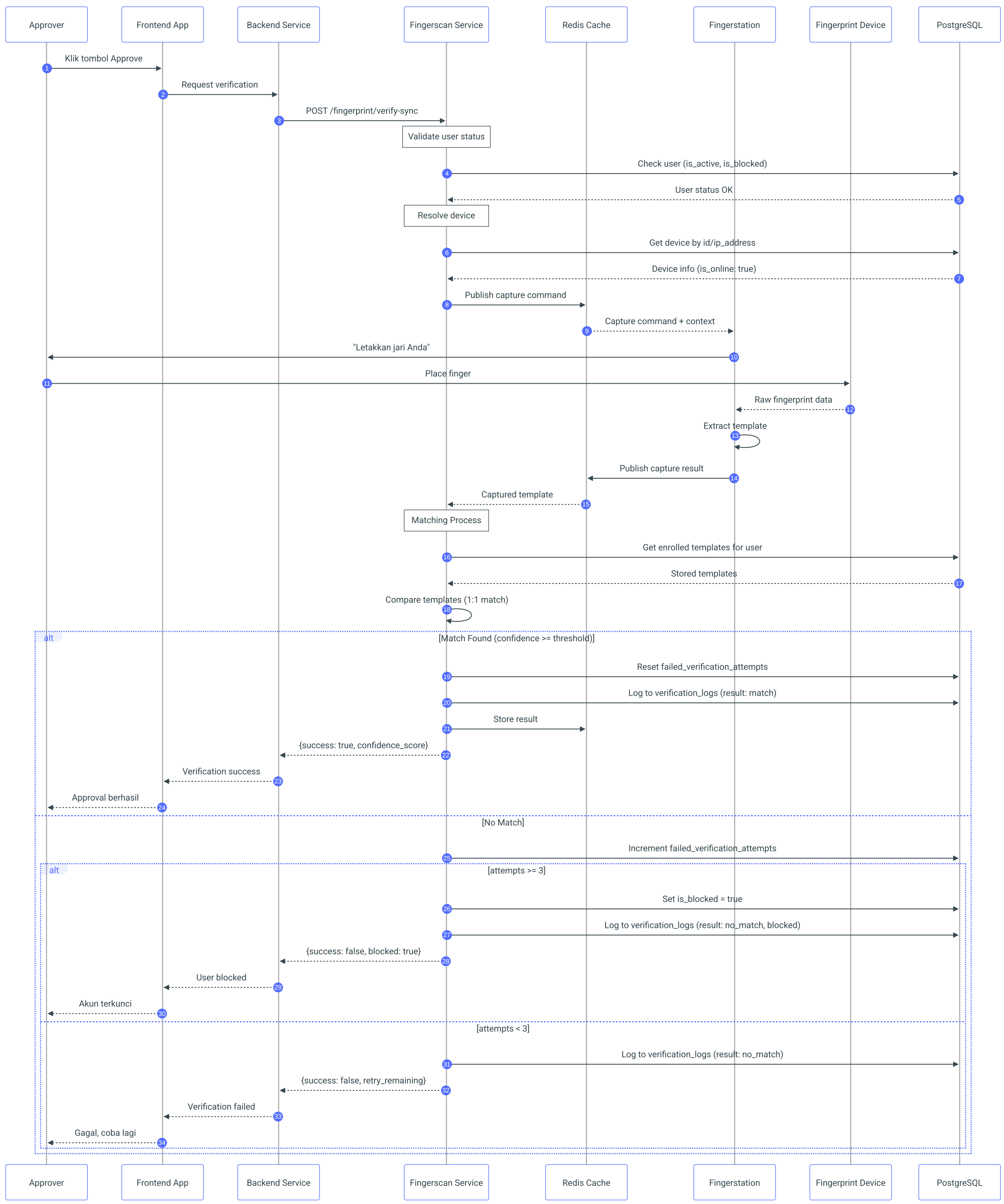
Penjelasan Alur:

Step	Dari	Ke	Deskripsi	Protokol
1-3	Admin → Service	Inisiasi enrollment session	REST API	
4-5	Service → DB	Buat record enrollment_session	SQL	
6-7	Service → Portal	Return session_id untuk tracking	HTTP Response	
8-10	Portal → Service	Request capture untuk setiap jari	REST API	

Step	Dari	Ke	Deskripsi	Protokol
11-12	Service → Station	Kirim capture command via Redis pub/sub	Redis	
13-15	Admin → Device	User meletakkan jari di scanner	Physical	
16-17	Station	Ekstrak template dari raw data	Internal	
18-19	Station → Service	Kirim template hasil capture via Redis	Redis	
20-21	Service → DB	Simpan template dan log enrollment	SQL	
22-25	Service → Admin	Return hasil capture	HTTP Response	
26-31	Portal → Admin	Complete enrollment dan konfirmasi	REST + UI	

Timeline Estimasi: - Step 1-7: ~200ms (session creation) - Step 8-25: ~5-10 detik per jari (capture + save) - Step 26-31: ~100ms (complete session) - **Total: ~16-31 detik** untuk 3 jari (user-dependent)

2.2.2 Verification Process Flow



Penjelasan Alur:

Step	Dari	Ke	Deskripsi	Protokol
1-3	Approver → Service	Request verifikasi saat approval transaksi	REST API	

Step	Dari	Ke	Deskripsi	Protokol
4-5	Service → DB	Validasi user aktif dan tidak di-block	SQL	
6-7	Service → DB	Resolve device dari device_id atau ip_address	SQL	
8-9	Service → Station	Kirim capture command via Redis pub/sub	Redis	
10-12	Approver → Device	User meletakkan jari di scanner	Physical	
13	Station	Ekstrak template dari raw fingerprint	Internal	
14-15	Station → Service	Kirim template via Redis	Redis	
16-17	Service → DB	Ambil enrolled templates untuk matching	SQL	
18	Service	Proses matching 1:1	Internal	
19-24	Service → App	Handle result (success/failed/blocked)	HTTP	

Timeline Estimasi: - Step 1-9: ~200ms (validation + command dispatch) - Step 10-15: ~2-5 detik (user action + capture) - Step 16-18: ~100ms (matching process) - Step 19-24: ~50ms (logging + response) - **Total: ~2.5-5.5 detik** (user-dependent)

Timeout Handling: - Sync mode timeout: 30 detik - Jika timeout tercapai, return HTTP 408 dengan status "timeout"

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3. Business Rules

ID	Category	Rule	Implementation
BR-001	Enrollment	Minimal 2 jari dari kedua tangan	Validation sebelum save
BR-002	Enrollment	Quality score minimal 40	Check saat capture
BR-003	Enrollment	Tidak boleh duplicate	Cross-check dengan existing templates
BR-004	Enrollment	User harus aktif di HRMIS	API call ke HRMIS
BR-005	Verification	Max retry 3x	Counter di database
BR-006	Verification	Timeout 30 detik	Session expiry
BR-007	Verification	Match threshold configurable	Default 40
BR-008	Verification	1 active session per user	Reject concurrent
BR-009	Lock	Auto lock setelah 3x gagal	System triggered
BR-010	Lock	Manual unlock by admin only	Role-based access
BR-011	Fallback	Approval dari Admin Kantor Pusat	Workflow approval
BR-012	Fallback	Max duration 30 hari	Validation
BR-013	Fallback	Auto disable setelah expired	Scheduler
BR-014	Audit	Semua aktivitas harus di-log	Aspect/interceptor

ID	Category	Rule	Implementation
BR-015	Audit	Retention minimal 2 tahun	Archival policy

4. Modul Sistem

4.1 Fingerstation (Client App)

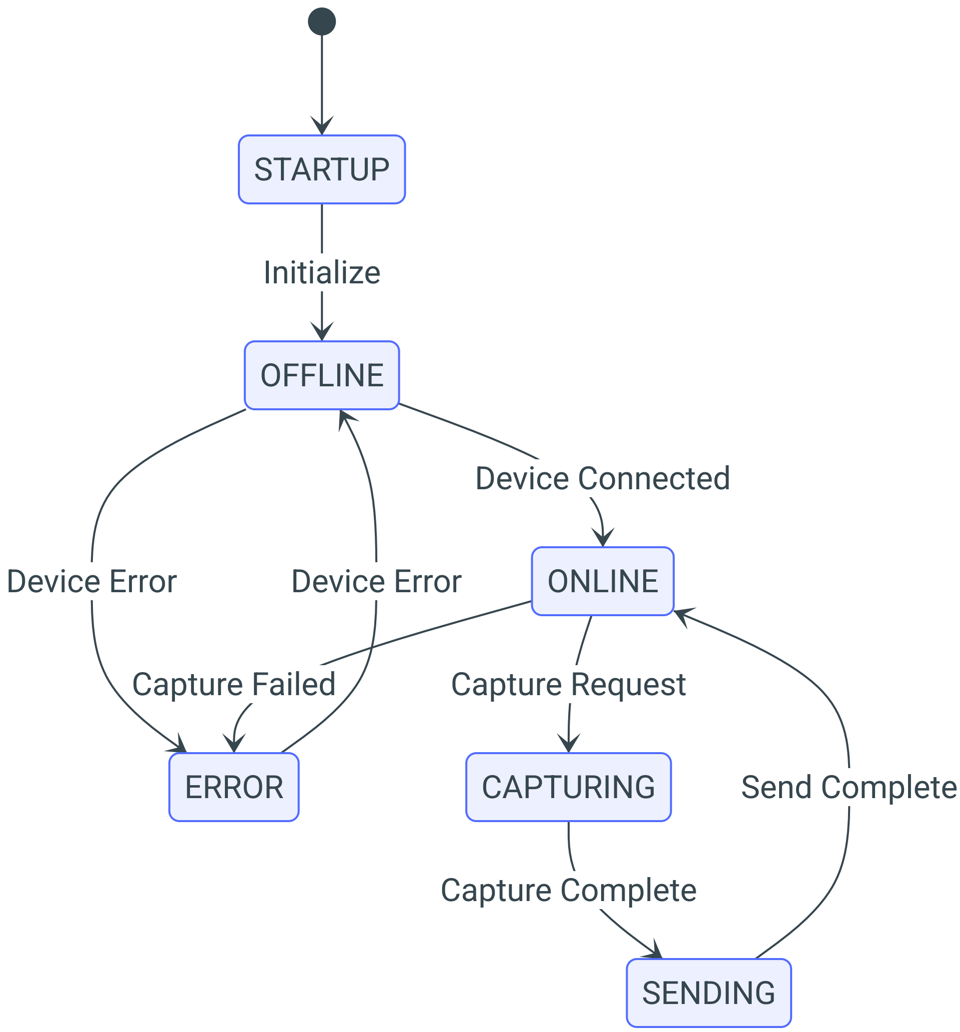
Deskripsi: Aplikasi desktop yang diinstal pada workstation cabang untuk menghubungkan device fingerprint dengan Fingerscan Service.

Teknologi: Electron / .NET Desktop App

Fungsi Utama:

No	Fungsi	Deskripsi
1	Device Connection	Mengelola koneksi dengan DigitalPersona 4500
2	Capture	Melakukan capture sidik jari dari device
3	Template Extraction	Mengekstrak template dari hasil capture
4	Secure Communication	Mengirim template ke service via WebSocket
5	Status Reporting	Melaporkan status device dan station

State Diagram:



4.2 Fingerscan Service (Backend)

Deskripsi: Backend service yang menangani logika bisnis verifikasi, enrollment, dan manajemen sistem.

Teknologi: Python Sanic (Async Framework)

Stack:

Component	Technology
Framework	Sanic 25.x (Async Python)
Database	PostgreSQL 15+

Component	Technology
ORM	SQLAlchemy 2.0 (Async)
Cache/Messaging	Redis (Pub/Sub)
Authentication	JWT + Bcrypt

Sub-modul (Services):

Sub-modul	File	Fungsi
Fingerprint Verification	fingerprint_verification_service.py	Matching template 1:1, logging hasil
Enrollment Service	enrollment_service.py	Session management, template capture
Device Service	device_service.py	Device registration, status, resolve
User Service	user_service.py	User CRUD, blocking, auth
Template Service	fingerprint_template_service.py	Template CRUD, upsert
Command Service	command_service.py	Dispatch command ke station via Redis
Redis Service	redis_service.py	Redis connection, pub/sub
Log Service	log_service.py	Audit logging ke database
Parameter Service	parameter_service.py	System configuration

4.3 Fingerscan Management (Web Portal)

Deskripsi: Portal web untuk administrasi sistem fingerprint.

Teknologi: Next.js / React/

Menu Struktur:

```
Fingerscan Management
├── Dashboard
│   ├── Station Status Overview
│   ├── Verification Statistics
│   └── Alert & Notifications
├── Enrollment
│   ├── New Enrollment
│   ├── Update Enrollment
│   └── Enrollment History
├── User Management
│   ├── User List
│   ├── Lock/Unlock User
│   └── Fallback Management
├── Device Management
│   ├── Station List
│   ├── Device Registration
│   └── Maintenance Schedule
├── Reports
│   ├── Verification Report
│   ├── Audit Trail
│   └── Usage Statistics
└── Settings
    ├── System Parameters
    ├── Branch Configuration
    └── Integration Settings
```

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5. Spesifikasi Fungsional

5.1 F01: Enrollment Sidik Jari

4.1.1 Deskripsi

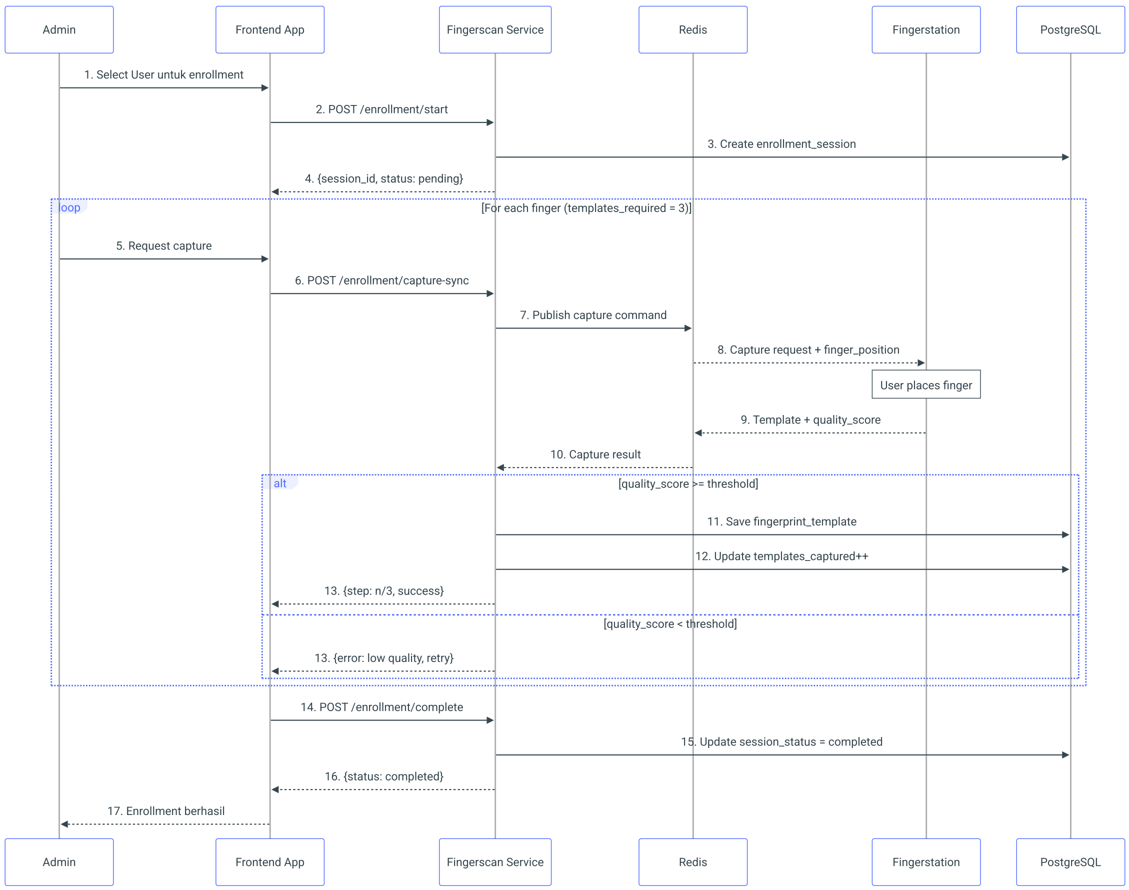
Proses pendaftaran sidik jari user ke dalam sistem untuk pertama kali. Enrollment dilakukan untuk user definitif (di Kantor Pusat) dan user alternate (di Cabang).

4.1.2 Precondition

- User terdaftar di sistem (exists in users table)
- User aktif (is_active = true)

- Device terdaftar dan online (is_online = true)
- Device dapat melakukan enrollment (can_enroll = true)

4.1.3 Flow Diagram



4.1.4 Business Rules

Rule ID	Rule	Validasi
BR-E01	Minimal 3 jari dari kedua tangan	System enforced
BR-E02	Template harus memenuhi quality threshold	Score >= 40
BR-E03	Tidak boleh duplicate dengan user lain	Matching check
BR-E04	User harus aktif di HRMIS	API validation

Rule ID	Rule	Validasi
BR-E05	Enrollment definitif hanya di Kantor Pusat	Location check

4.1.5 Data Input

Field	Type	Required	Validation
user_id	String	Yes	Exists in HRMIS, Active
finger_index	Integer	Yes	1-10 (mapping jari)
template_data	Binary	Yes	Quality >= 40
station_id	UUID	Yes	Station ONLINE
enrolled_by	String	Yes	From session

4.1.6 Data Output

Field	Type	Description
enrollment_id	UUID	ID enrollment
status	Enum	SUCCESS, FAILED
finger_count	Integer	Jumlah jari terdaftar
message	String	Status message

4.2 F02: Verifikasi Sidik Jari

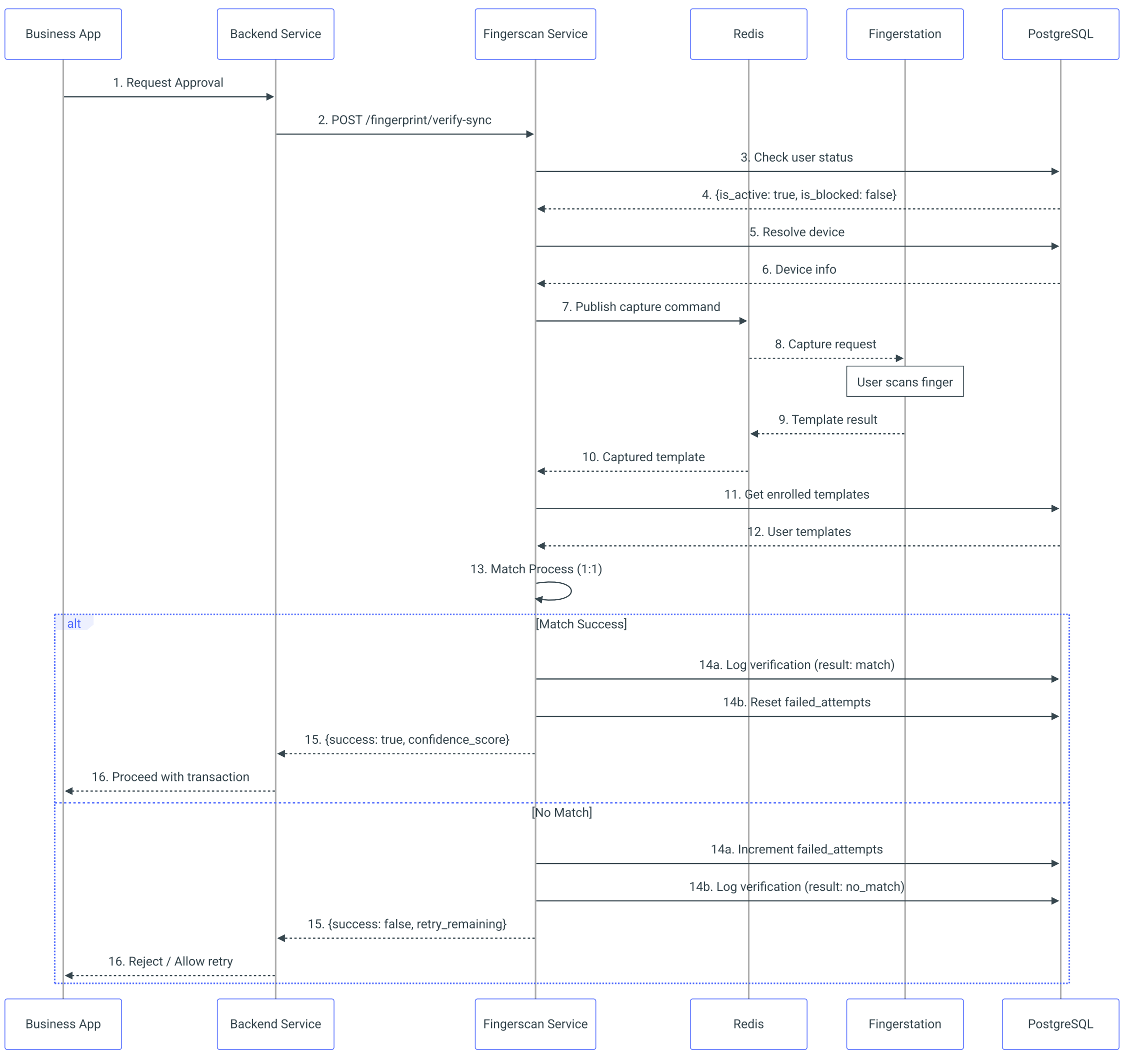
4.2.1 Deskripsi

Proses verifikasi sidik jari user saat melakukan approval transaksi. Verifikasi dipicu oleh aplikasi bisnis (DAF, BDS, dll) ketika user melakukan aksi approval.

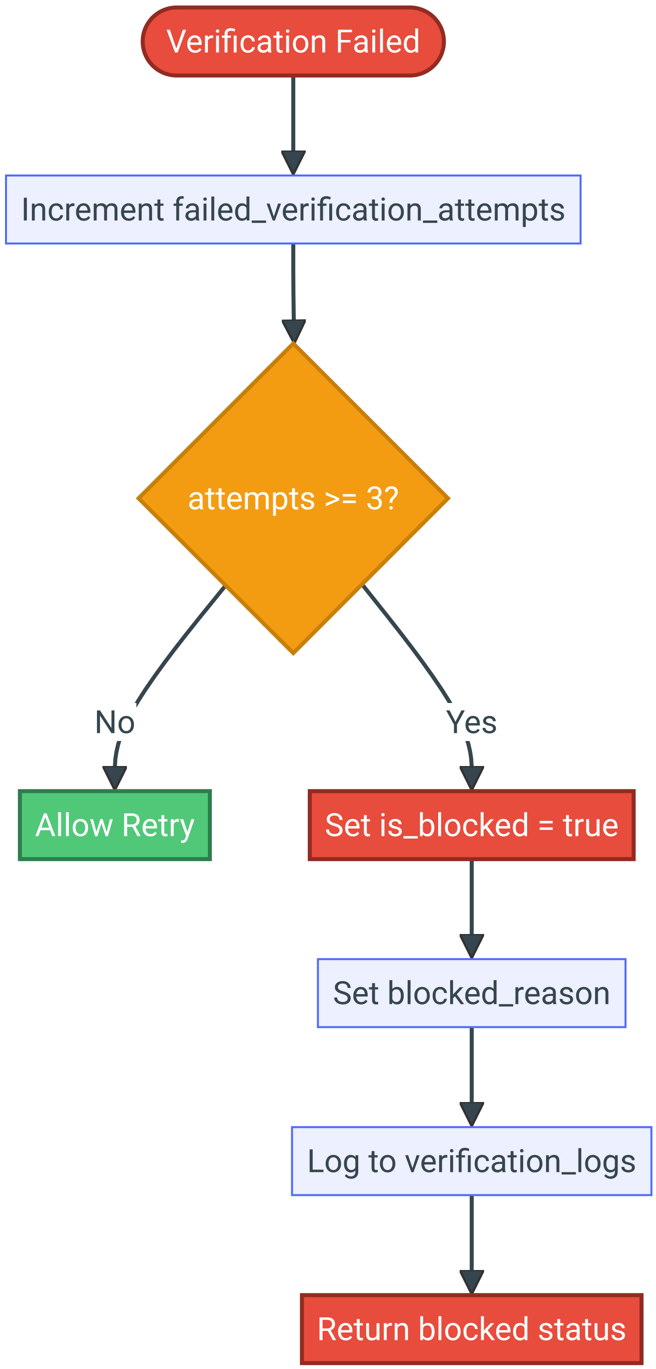
4.2.2 Precondition

- User telah melakukan enrollment (memiliki fingerprint_templates)
- User tidak dalam status blocked (is_blocked = false)
- User aktif (is_active = true)
- Device dalam status online (is_online = true)

4.2.3 Sequence Diagram



4.2.4 Flow: Verifikasi Gagal



4.2.5 Business Rules

Rule ID	Rule	Action
BR-V01	Max retry 3x	Set is_blocked=true setelah 3x gagal

Rule ID	Rule	Action
BR-V02	Timeout 30 detik	Return HTTP 408 jika timeout
BR-V03	Match threshold configurable	Via parameters table
BR-V04	User harus aktif	Check is_active sebelum verify
BR-V05	User tidak boleh blocked	Check is_blocked sebelum verify

4.2.6 API Request

```
POST /fingerprint/verify-sync
{
  "user_id": "550e8400-e29b-41d4-a716-446655440000",
  "device_id": "660e8400-e29b-41d4-a716-446655440001",
  "username": "john.doe",
  "application": "DAF-CORE",
  "transaction_code": "TRX20250128001",
  "supervisor_name": "Jane Smith",
  "reference_number": "REF-001",
  "description": "Transfer approval Rp 50.000.000"
}
```

4.2.7 API Response

Success:

```
{
  "status": "success",
  "result": {
    "match": true,
    "confidence_score": 95.2
  },
  "timestamp": "2025-01-28T10:30:00Z",
  "verification_id": "verify_sync_660e8400_1706438400",
  "device_id": "660e8400-e29b-41d4-a716-446655440001"
}
```

Failed (Blocked):

```
{
  "status": "blocked",
  "message": "Too many failed verification attempts",
```

```json
  "user_id": "550e8400-e29b-41d4-a716-446655440000"
}
```


Timeout:


```json
{
  "status": "timeout",
  "message": "Verification process timed out. Please try again.",
  "verification_id": "verify_sync_660e8400_1706438400"
}
```

4.3 F03: Manajemen Lock/Unlock

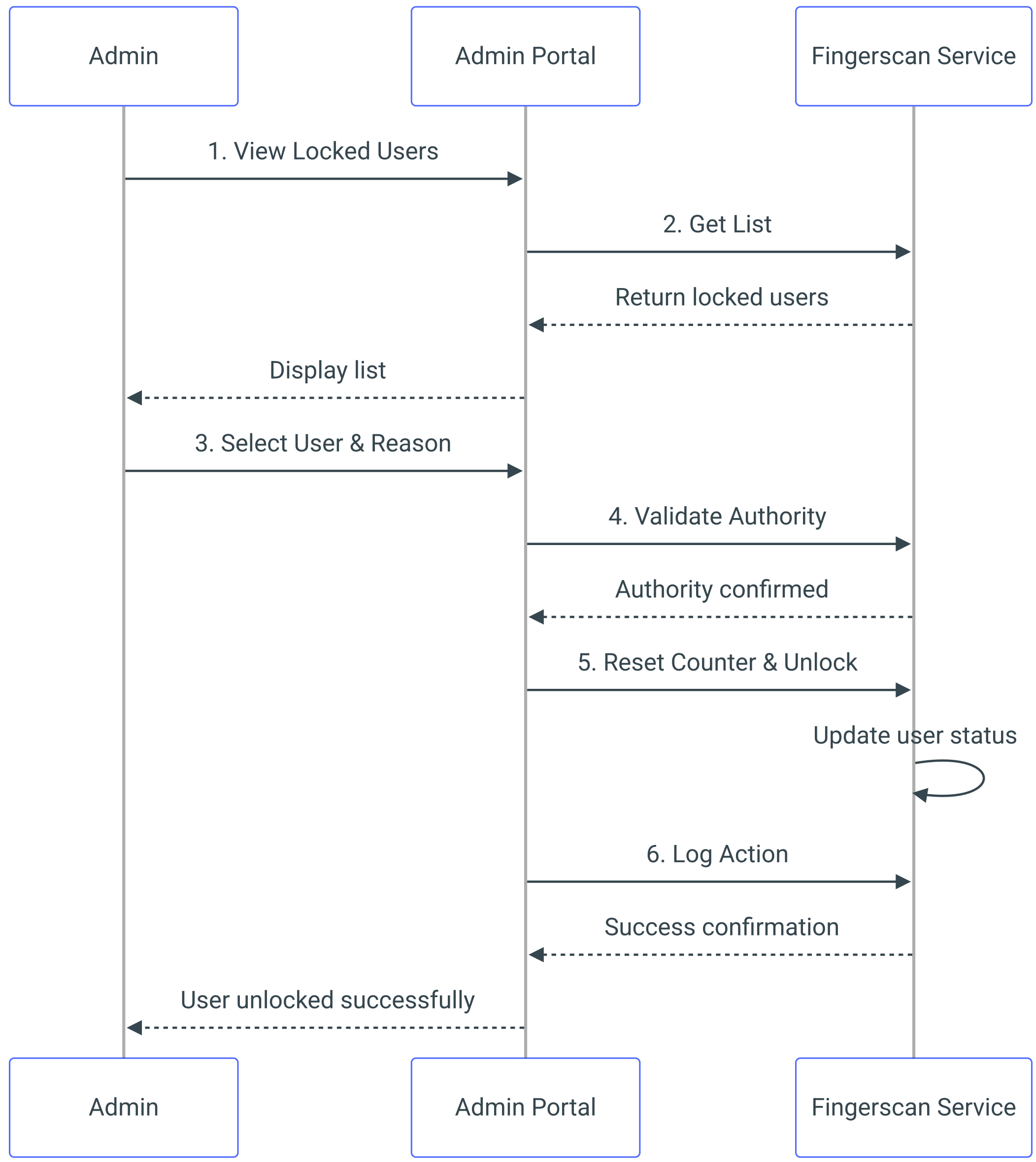
4.3.1 Deskripsi

Fungsi untuk mengunci dan membuka kunci akun user di level fingerprint.

4.3.2 Kondisi Lock

Kondisi	Trigger	Auto/Manual
3x verifikasi gagal	System	Auto
Request dari admin	Admin action	Manual
User resign/non-aktif	HRMIS sync	Auto
Security incident	Security team	Manual

4.3.3 Proses Unlock



4.4 F04: Fallback Password

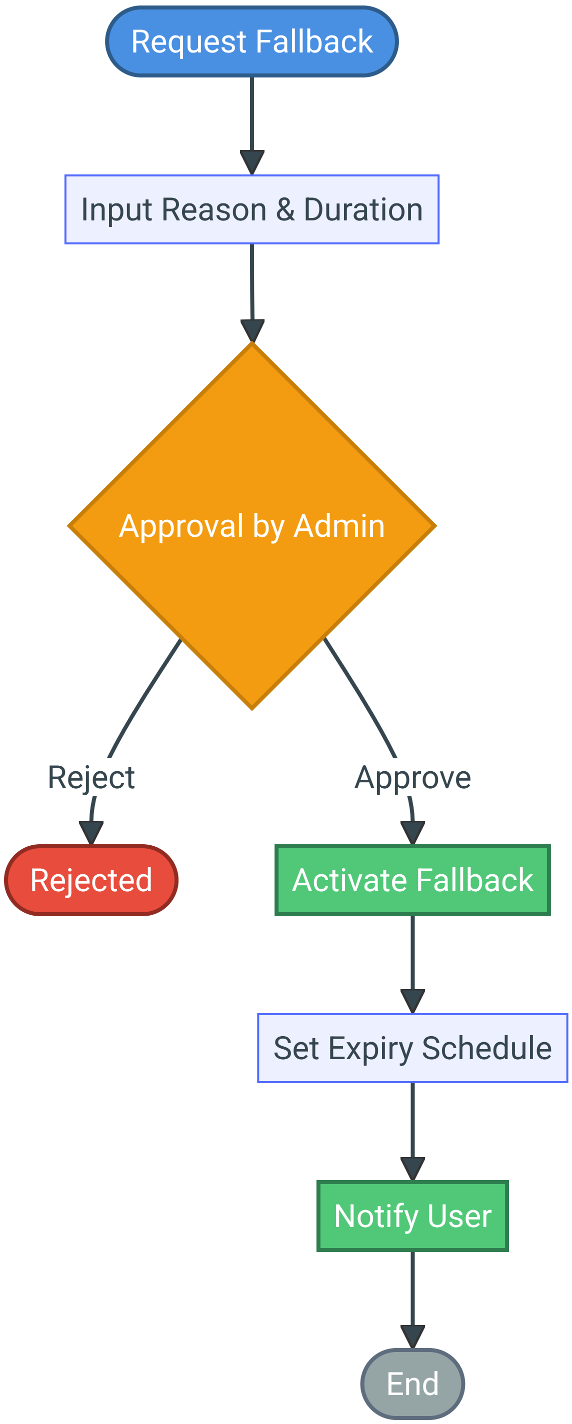
4.4.1 Deskripsi

Mekanisme cadangan yang memungkinkan user melakukan approval dengan password ketika fingerprint tidak dapat digunakan (device rusak, kondisi darurat).

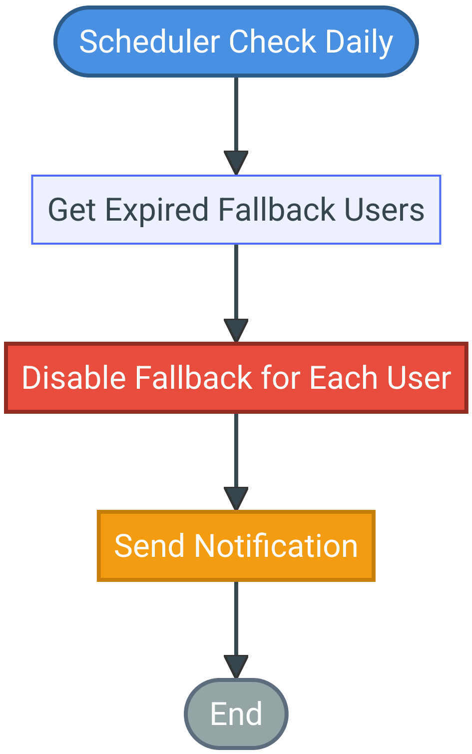
4.4.2 Ketentuan Fallback

Parameter	Nilai	Keterangan
Approval Required	Admin Kantor Pusat (TI)	Mandatory
Max Duration	30 hari	Configurable
Min Duration	1 hari	Configurable
Auto Expire	Yes	Scheduler based
Reason Required	Yes	Mandatory field

4.4.3 Flow Aktivasi



4.4.4 Flow Expiry



4.5 F05: Manajemen Device & Station

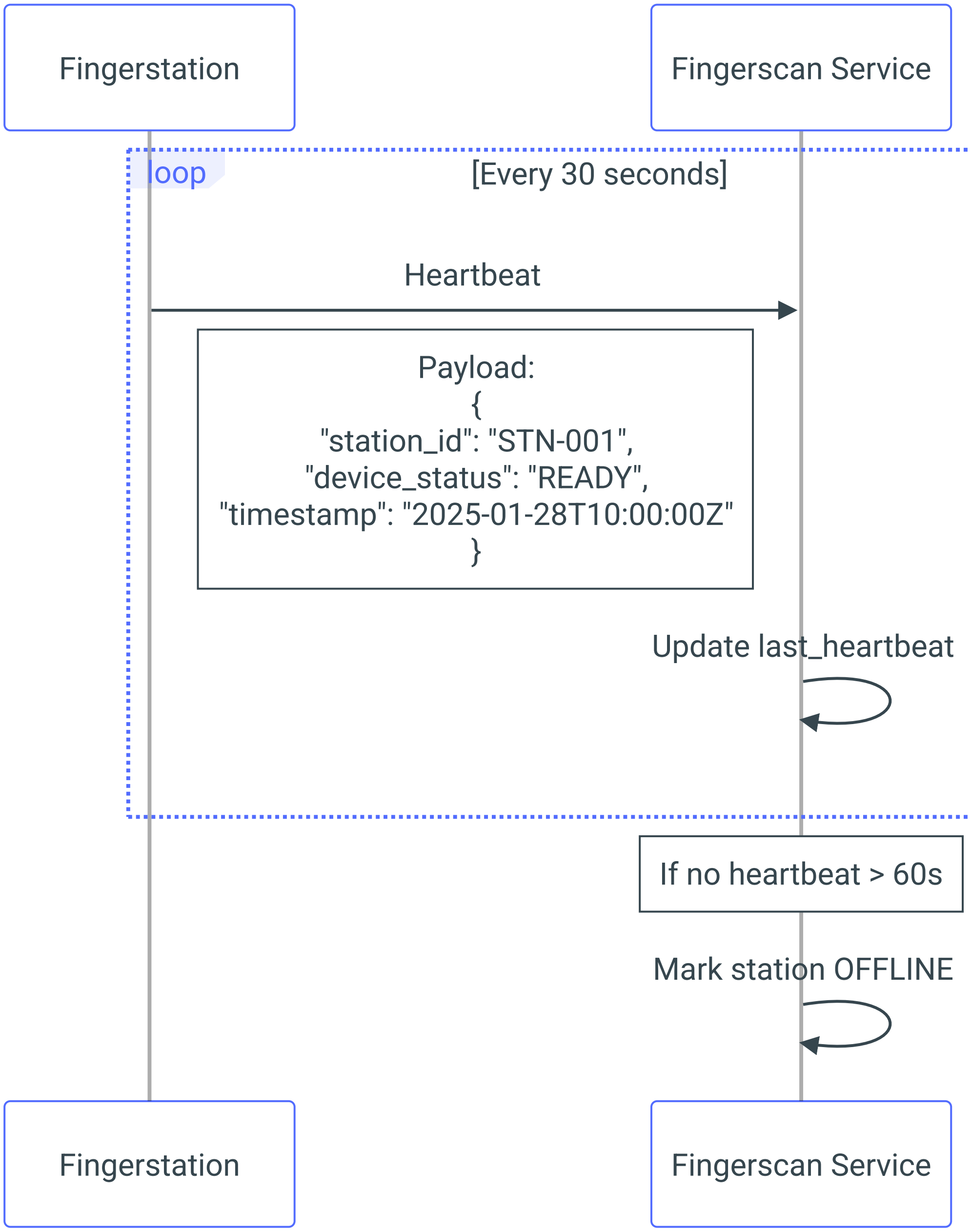
4.5.1 Registrasi Station

Field	Type	Required	Description
station_name	String	Yes	Nama identifikasi station
branch_code	String	Yes	Kode cabang
device_serial	String	Yes	Serial number device
ip_address	String	Yes	IP address station
location	String	No	Lokasi fisik (Teller 1, CS, dll)

4.5.2 Status Monitoring

Status	Kondisi	Action
ONLINE	Connected & device ready	Available for use
OFFLINE	No heartbeat > 60s	Alert to admin
ERROR	Device malfunction	Require maintenance
MAINTENANCE	Scheduled maintenance	Temporarily unavailable

4.5.3 Heartbeat Mechanism



4.6 F06: Audit & Reporting

4.6.1 Data Audit Trail

Field	Type	Description
log_id	UUID	Unique log identifier

Field	Type	Description
timestamp	DateTime	Waktu event
event_type	Enum	ENROLLMENT, VERIFICATION, LOCK, UNLOCK, FALLBACK
user_id	String	User yang terlibat
actor_id	String	User yang melakukan aksi
application_code	String	Kode aplikasi source
transaction_ref	String	Referensi transaksi
transaction_type	String	Jenis transaksi
station_id	String	Station yang digunakan
ip_address	String	IP address
result	Enum	SUCCESS, FAILED
details	JSON	Additional details

4.6.2 Report Types

Report	Deskripsi	Retention
Verification Summary	Ringkasan verifikasi harian/bulanan	2 tahun
Failed Verification	Daftar verifikasi gagal	2 tahun
User Activity	Aktivitas per user	2 tahun
Station Usage	Penggunaan per station	1 tahun

Report	Deskripsi	Retention
Fallback Usage	Penggunaan fallback	2 tahun

5. Spesifikasi Antarmuka

5.1 API Specification

5.1.1 Base URL

```
Production: https://fingerscan.bank.internal/api
Staging:    https://fingerscan-stg.bank.internal/api
```

5.1.2 Technology Stack

Component	Technology
Framework	Sanic (Async Python)
Database	PostgreSQL 15+
ORM	SQLAlchemy 2.0 (Async)
Cache/Messaging	Redis (Pub/Sub)
Authentication	JWT (24h expiration)
Real-time	WebSocket
API Documentation	OpenAPI 3.0 / Swagger

5.1.3 Authentication

Semua API menggunakan JWT Bearer Token:

```
Authorization: Bearer <access_token>
```

Endpoint	Method	Description
/auth/register	POST	User registration
/auth/login	POST	Login dan mendapatkan JWT token
/auth/me	GET	Get current user info

5.1.4 Endpoints

Device Management (`/devices`)

Method	Endpoint	Description
GET	/devices	List semua devices (dengan filter)
GET	/devices/pending	List devices pending approval
POST	/devices	Register device baru
GET	/devices/{device_id}	Get device detail
PUT	/devices/{device_id}	Update device
DELETE	/devices/{device_id}	Hapus device
POST	/devices/{device_id}/approve	Approve pending device
POST	/devices/{device_id}/status	Check device online status

Enrollment (`/enrollment`)

Method	Endpoint	Description
POST	/enrollment/start	Mulai enrollment session

Method	Endpoint	Description
POST	/enrollment/capture	Request capture (async)
POST	/enrollment/capture-sync	Request capture (sync, 15s timeout)
POST	/enrollment/complete	Complete enrollment session
GET	/enrollment/sessions	List enrollment sessions
GET	/enrollment/sessions/{session_id}	Get session detail
GET	/enrollment/result/{session_id}	Get capture result
PUT	/enrollment/sessions/{session_id}/template	Add template to session

Fingerprint Verification & Identification (`/fingerprint`)

Method	Endpoint	Description
POST	/fingerprint/verify	Start verification (async)
POST	/fingerprint/verify-sync	Verify synchronous (timeout)
GET	/fingerprint/verification/result/{id}	Get verification result
POST	/fingerprint/identify	Start identification 1:N (async)
POST	/fingerprint/identify-sync	Identify synchronous
GET	/fingerprint/identification/result/{id}	Get identification result

Template Management (`/fingerprint/templates`)

Method	Endpoint	Description
GET	/fingerprint/templates	List templates (dengan filter)
GET	/fingerprint/templates/{template_id}	Get template detail
POST	/fingerprint/templates	Create/update template
PUT	/fingerprint/templates/{template_id}	Update template
DELETE	/fingerprint/templates/{template_id}	Hapus template

User Management (`/users`)

Method	Endpoint	Description
GET	/users	List users (dengan filter)
GET	/users/{user_id}	Get user detail
PUT	/users/{user_id}	Update user (termasuk unblock)
DELETE	/users/{user_id}	Hapus user

Configuration (`/parameters`)

Method	Endpoint	Description
GET	/parameters	List configuration parameters
GET	/parameters/{parameter_id}	Get parameter
POST	/parameters	Create parameter
PUT	/parameters/{parameter_id}	Update parameter

Method	Endpoint	Description
DELETE	/parameters/{parameter_id}	Delete parameter

5.1.5 Contoh API Detail

POST /fingerprint/verify-sync (Verifikasi Synchronous)

Request:


```json
{
  "user_id": "550e8400-e29b-41d4-a716-446655440000",
  "device_id": "660e8400-e29b-41d4-a716-446655440001",
  "username": "john.doe",
  "application": "DAF-CORE",
  "supervisor_name": "Jane Smith",
  "reference_number": "TRX20250128001",
  "description": "Transfer approval"
}
```


Response (Success):


```json
{
  "success": true,
  "confidence_score": 95.2,
  "message": "Fingerprint verified successfully",
  "verification_id": "770e8400-e29b-41d4-a716-446655440002"
}
```


Response (Error - 408 Timeout):


```json
{
  "error": "Verification timeout",
  "message": "No fingerprint captured within timeout period"
}
```


POST /enrollment/start (Mulai Enrollment)

Request:


```json
{
  "user_id": "550e8400-e29b-41d4-a716-446655440000",
  "device_id": "660e8400-e29b-41d4-a716-446655440001",
  "templates_required": 3,
```

```
  "ip_address": "192.168.1.100"
}
```

Response (201):

```
{
  "status": "success",
  "id": "880e8400-e29b-41d4-a716-446655440003",
  "user_id": "550e8400-e29b-41d4-a716-446655440000",
  "device_id": "660e8400-e29b-41d4-a716-446655440001",
  "session_status": "pending",
  "templates_captured": 0,
  "templates_required": 3
}
```

5.2 WebSocket Events

5.2.1 Connection

```
wss://fingerscan.bank.internal/ws/station
```

5.2.2 Communication Pattern

Sistem menggunakan Redis Pub/Sub untuk komunikasi async antara API dan Fingerstation:

1. **API** mengirim command ke Redis channel
2. **Fingerstation** subscribe ke channel dan menerima command
3. **Fingerstation** mengirim hasil capture ke Redis
4. **API** mengambil hasil dari Redis (polling atau callback)

5.2.3 Events

Event	Direction	Payload
station.connect	Station → Service	{ station_id, device_serial }
station.heartbeat	Station → Service	{ station_id, device_status, is_online }
capture.request	Service → Station	{ session_id, finger_position, timeout }
capture.result	Station → Service	{ session_id, template_data, quality_score }

Event	Direction	Payload
capture.error	Station → Service	{ session_id, error_code, message }
echo.request	Service → Station	{ command_id, echo_text }
echo.result	Station → Service	{ command_id, result }
station.disconnect	Station → Service	{ station_id, reason }

5.3 External System Integration

5.3.1 OIDC Integration

Parameter	Value
Authorization Endpoint	https://sso.bank.internal/oauth/authorize
Token Endpoint	https://sso.bank.internal/oauth/token
Userinfo Endpoint	https://sso.bank.internal/oauth/userinfo
Scopes	openid profile email

5.3.2 HRMIS Integration

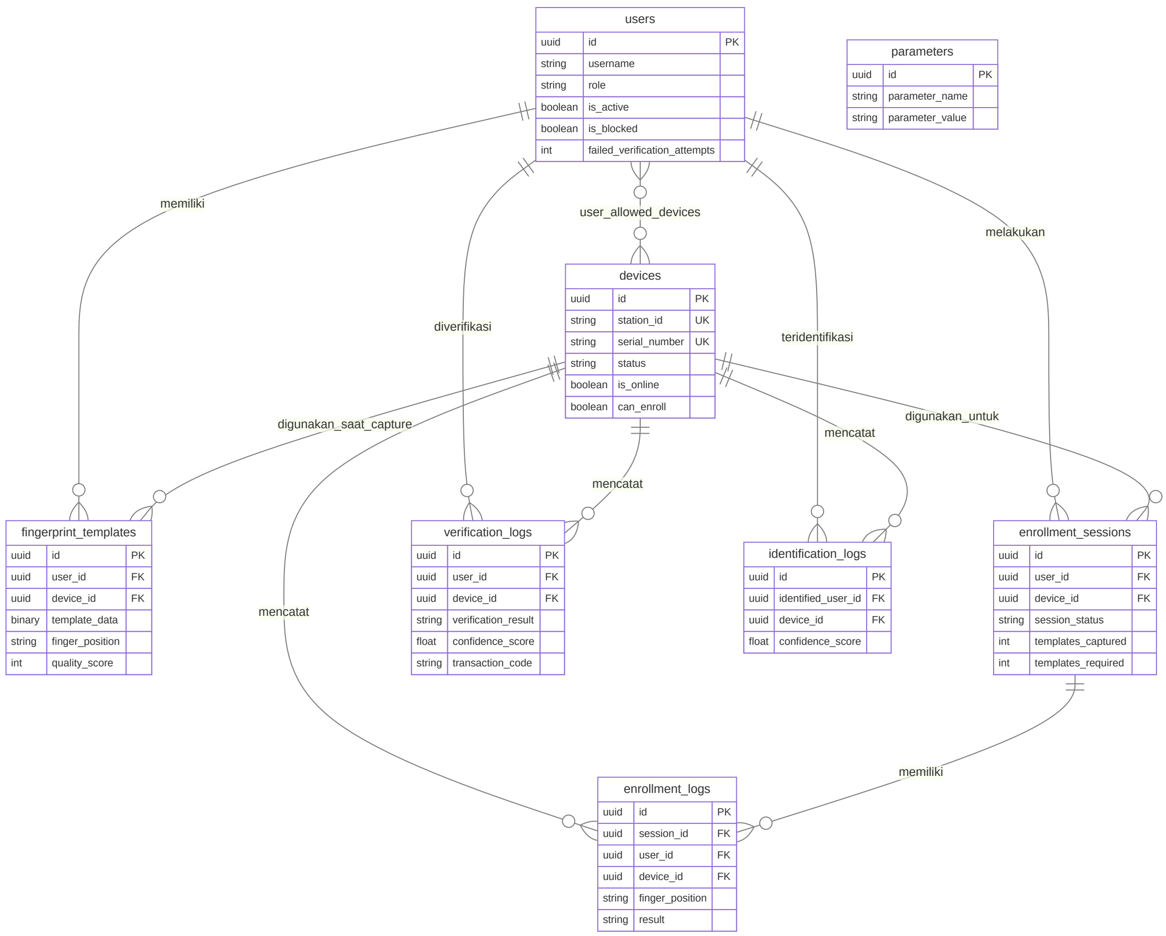
Endpoint	Method	Purpose
/api/employee/{id}	GET	Validate employee status
/api/employee/{id}/branch	GET	Get employee branch

---

6. Spesifikasi Data

6.1 Entity Relationship Diagram

> [!NOTE] Diagram berikut menunjukkan relasi antar tabel. Detail field lengkap akan didokumentasikan dalam TSD.



6.2 Deskripsi Tabel Database

> [!NOTE] Detail lengkap spesifikasi kolom, tipe data, constraint, dan index akan didokumentasikan dalam TSD (Technical Specification Document).

6.2.1 users

Tabel master untuk menyimpan data user yang ter-enroll dalam sistem biometrik. Menyimpan informasi kredensial, role (user/admin), status aktif, counter failed verification untuk auto-lock, dan informasi blocking.

6.2.2 devices

Tabel master station/device fingerprint scanner. Menyimpan informasi station_id, serial_number, manufacturer, model, IP address, status koneksi (is_online), kemampuan enrollment (can_enroll), dan approval status (pending/approved/active).

6.2.3 fingerprint_templates

Tabel untuk menyimpan template sidik jari user. Setiap user dapat memiliki multiple records (maksimal 10 jari dengan unique constraint pada kombinasi user_id + finger_position). Template disimpan dalam format base64. Menyimpan quality_score untuk validasi kualitas capture.

6.2.4 enrollment_sessions

Tabel untuk mencatat sesi enrollment. Menyimpan progress enrollment (templates_captured vs templates_required), status session (pending/in_progress/completed/failed), dan referensi ke user dan device yang digunakan.

6.2.5 verification_logs

Tabel transaksi untuk mencatat setiap proses verifikasi fingerprint 1:1. Menyimpan hasil matching (match/no_match/error), confidence_score, referensi ke aplikasi dan transaksi, serta informasi audit (IP address, timestamp).

6.2.6 identification_logs

Tabel transaksi untuk mencatat proses identifikasi 1:N. Menyimpan user yang teridentifikasi, confidence_score, dan informasi audit.

6.2.7 enrollment_logs

Tabel detail untuk mencatat setiap capture dalam sesi enrollment. Menyimpan finger_position, quality_score, hasil capture (success/failure), dan pesan error jika ada.

6.2.8 user_allowed_devices (Junction Table)

Tabel many-to-many untuk mengatur device mana saja yang boleh digunakan oleh user tertentu. Menyimpan role assignment dan status aktif.

6.2.9 parameters

Tabel konfigurasi sistem untuk menyimpan parameter-parameter seperti threshold matching, timeout, dan konfigurasi lainnya.

---

7. Spesifikasi User Interface

7.1 Fingerstation UI

TBD

7.3 Management Portal

TBD <!-- **Dashboard:**

```
┌─────────────────────────────────────────────────────────────────────┐
│  Fingerscan Management                      [User: Admin] [Logout]  │
├──────────────────┬──────────────────────────────────────────────────┤
│                  │                                                  │
│  Dashboard       │  ┌───────────────────────────────────────────┐  │
│  Enrollment      │  │  Station Status                           │  │
│  User Mgmt       │  ├───────────────────────────────────────────┤  │
│  Device Mgmt     │  │  ● Online: 45     ○ Offline: 3            │  │
│  Reports         │  │  ⚠ Error: 2      🔧 Maintenance: 1        │  │
│  Settings        │  └───────────────────────────────────────────┘  │
│                  │                                                  │
│                  │  ┌───────────────────────────────────────────┐  │
│                  │  │  Today's Statistics                       │  │
│                  │  ├───────────────────────────────────────────┤  │
│                  │  │  Total Verifications: 1,234               │  │
│                  │  │  Success Rate: 98.5%                      │  │
│                  │  │  Avg Response Time: 2.3s                  │  │
│                  │  │  Failed Attempts: 19                      │  │
│                  │  │  Users Locked: 2                          │  │
│                  │  └───────────────────────────────────────────┘  │
│                  │                                                  │
│                  │  ┌───────────────────────────────────────────┐  │
│                  │  │  Recent Alerts                            │  │
│                  │  ├───────────────────────────────────────────┤  │
│                  │  │  ⚠ Station STN-005 offline (5 min ago)    │  │
│                  │  │  🔒 User USR045 locked (10 min ago)       │  │
│                  │  │  ⚠ Station STN-012 device error           │  │
│                  │  └───────────────────────────────────────────┘  │
│                  │                                                  │
└──────────────────┴──────────────────────────────────────────────────┘
``` -->

---

8. Business Rules

ID	Category	Rule	Implementation
BR-001	Enrollment	Minimal 2 jari dari kedua tangan	Validation sebelum save
BR-002	Enrollment	Quality score minimal 40	Check saat capture
BR-003	Enrollment	Tidak boleh duplicate	Cross-check dengan existing
templates
```
```

````
| BR-004 | Enrollment | User harus aktif di HRMIS | API call ke HRMIS |
| BR-005 | Verification | Max retry 3x | Counter di database |
| BR-006 | Verification | Timeout 30 detik | Session expiry |
| BR-007 | Verification | Match threshold configurable | Default 40 |
| BR-008 | Verification | 1 active session per user | Reject concurrent |
| BR-009 | Lock | Auto lock setelah 3x gagal | System triggered |
| BR-010 | Lock | Manual unlock by admin only | Role-based access |
| BR-011 | Fallback | Approval dari Admin Kantor Pusat | Workflow approval |
| BR-012 | Fallback | Max duration 30 hari | Validation |
| BR-013 | Fallback | Auto disable setelah expired | Scheduler |
| BR-014 | Audit | Semua aktivitas harus di-log | Aspect/interceptor |
| BR-015 | Audit | Retention minimal 2 tahun | Archival policy |

---

## 9. Error Handling

### 9.1 Error Codes

| Code | Category | Message | Action |
|------|----------|---------|--------|
| E001 | Enrollment | User not found in HRMIS | Verify user data |
| E002 | Enrollment | User already enrolled | Use update instead |
| E003 | Enrollment | Low quality capture | Retry capture |
| E004 | Enrollment | Duplicate template detected | Investigate |
| E005 | Enrollment | Station not available | Check station status |
| E101 | Verification | User not enrolled | Complete enrollment first |
| E102 | Verification | User is locked | Contact admin |
| E103 | Verification | Session timeout | Retry verification |
| E104 | Verification | No match found | Retry or use fallback |
| E105 | Verification | Station offline | Use different station |
| E106 | Verification | Concurrent session exists | Wait and retry |
| E201 | Fallback | Not authorized | Contact IT admin |
| E202 | Fallback | Invalid duration | Check parameters |
| E203 | Fallback | Already active | Check existing fallback |
| E301 | Station | Device not connected | Check USB connection |
| E302 | Station | Device error | Restart device |
| E303 | Station | Network error | Check connectivity |
| E901 | System | Internal server error | Contact support |
| E902 | System | Database error | Contact support |
| E903 | System | Service unavailable | Try again later |

### 9.2 Error Response Format

```json
{
  "code": "E104",
  "message": "No match found",
  "details": {
    "session_id": "550e8400-e29b-41d4-a716-446655440000",
    "retry_remaining": 2
  },
````
````

```
  "timestamp": "2025-01-28T10:30:00Z"
}
```

---

10. Security Specification

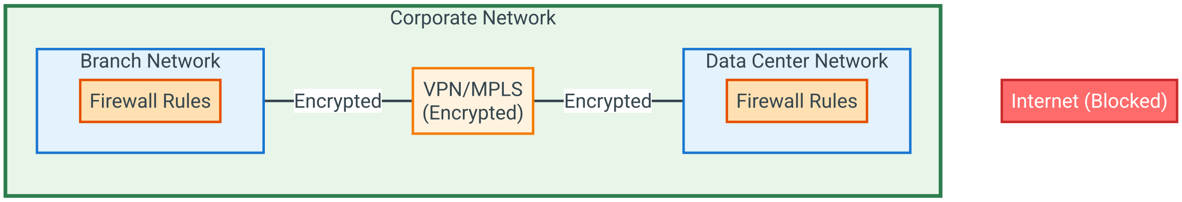
10.1 Data Security

Aspect	Specification
Template Storage	AES-256 encryption at rest
Data Transmission	TLS 1.3 + mTLS
Key Management	HSM-based key storage
Template Format	ISO/IEC 19794-2 (tidak menyimpan gambar)

10.2 Access Control

Role	Permissions
Teller/CS	View own verification status
Approver	Perform verification
Branch Admin	Enrollment alternate, view station
IT Admin	Full access except audit deletion
Auditor	Read-only audit logs

10.3 Network Security



10.4 Audit Trail

Semua akses dan operasi dicatat dengan informasi:

- Timestamp (dengan timezone)
- User ID dan IP address
- Action performed
- Result (success/failure)
- Request/Response payload (sanitized)

---

11. Performance Requirements

Metric	Target	Measurement
Verification Response Time	< 5 detik (P95)	End-to-end
Enrollment Response Time	< 10 detik per jari	End-to-end
API Response Time	< 200ms (P95)	Server-side
Concurrent Verifications	100 per detik	Load test
System Availability	99.9%	Uptime monitoring
Database Query Time	< 50ms (P95)	Query profiling
WebSocket Latency	< 100ms	Network monitoring

---

12. Lampiran

Lampiran A: Finger Position Mapping

Position (String)	Tangan	Jari
right_thumb	Kanan	Jempol
right_index	Kanan	Telunjuk
right_middle	Kanan	Tengah
right_ring	Kanan	Manis
right_little	Kanan	Kelingking
left_thumb	Kiri	Jempol
left_index	Kiri	Telunjuk
left_middle	Kiri	Tengah
left_ring	Kiri	Manis
left_little	Kiri	Kelingking

> **Note:** Setiap user hanya dapat memiliki satu template per finger_position (unique constraint pada user_id + finger_position).

Lampiran B: Status Transition

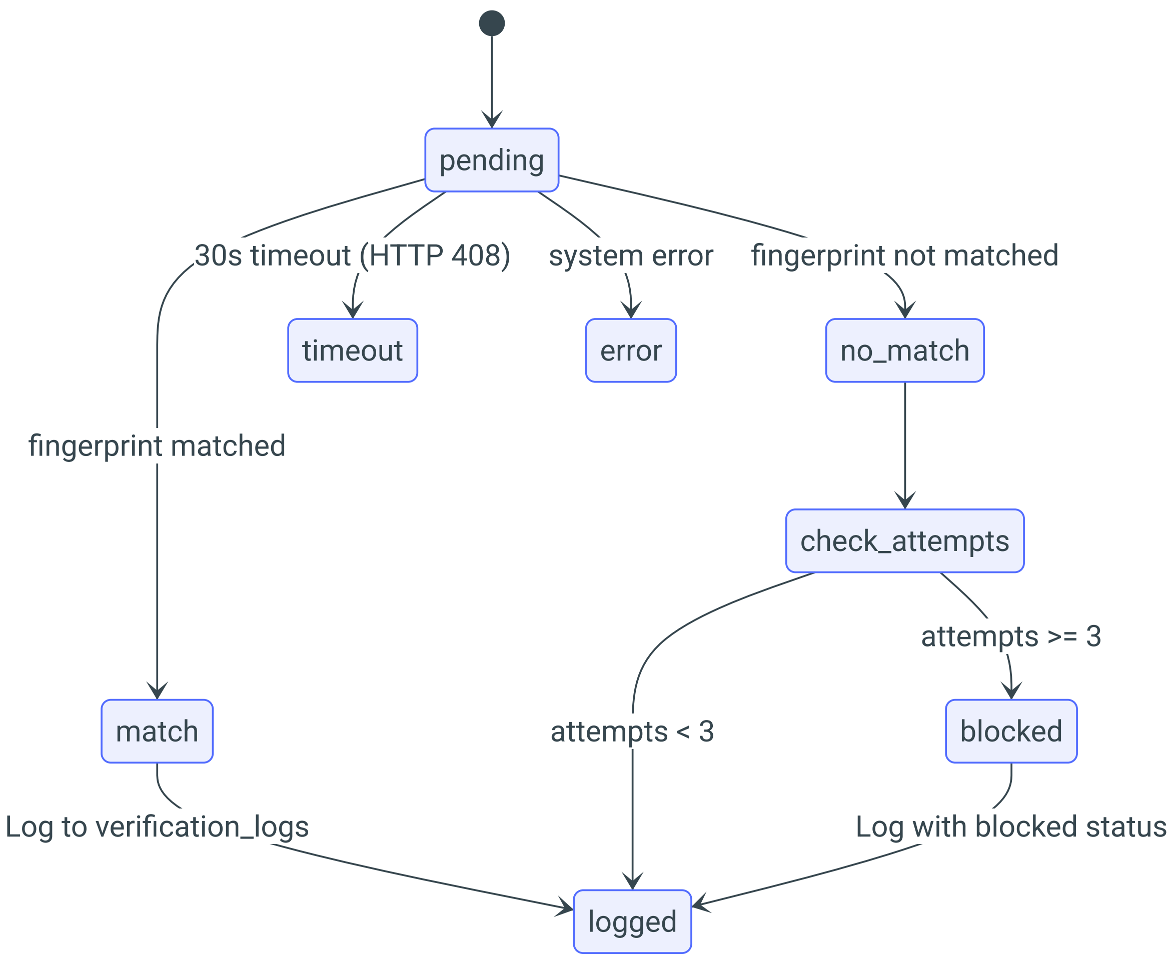
User Status:

```
stateDiagram-v2
    [*] --> is_active_true
    is_active_true --> is_blocked_true: 3x failed_verification_attempts
    is_blocked_true --> is_active_true: PUT /users/{id} (unblock: true)
    is_active_true --> is_active_false: Admin deactivate
    is_active_false --> is_active_true: Admin activate
```

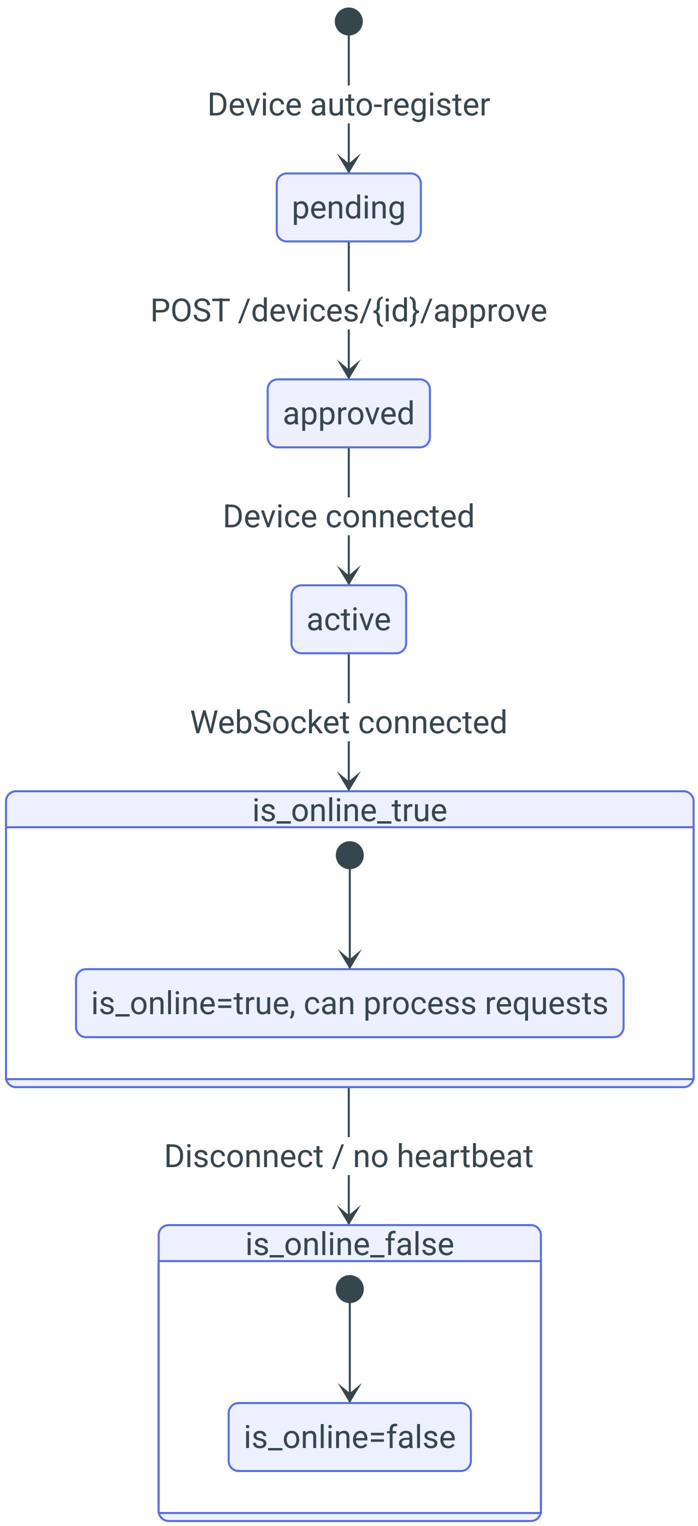
```
    state is_active_true {
        [*] --> Normal
        Normal: is_active=true, is_blocked=false
    }

    state is_blocked_true {
        [*] --> Blocked
        Blocked: is_blocked=true, blocked_reason set
    }
```

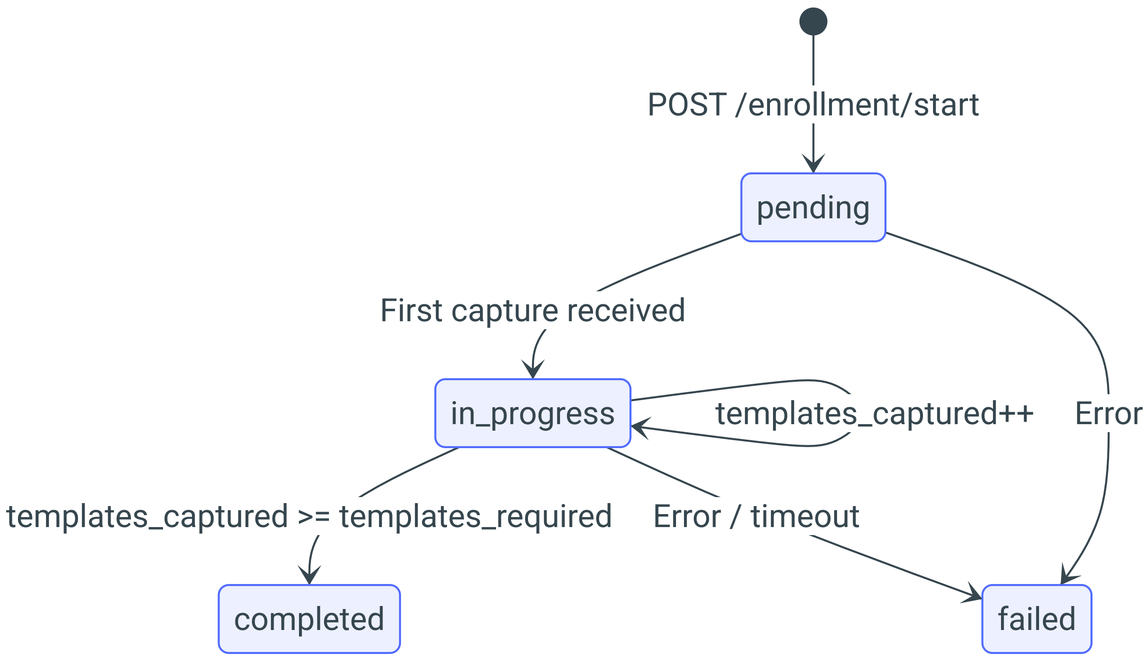
Verification Result:



Device Status:



Enrollment Session Status:



Lampiran C: Checklist Deployment

- [ ] Database schema created
- [ ] Redis cluster configured
- [ ] OIDC client registered
- [ ] mTLS certificates installed
- [ ] Firewall rules configured
- [ ] Monitoring setup complete
- [ ] Backup procedures tested
- [ ] Disaster recovery plan documented
- [ ] User training completed
- [ ] UAT sign-off obtained

Riwayat Perubahan

Versi	Tanggal	Penulis	Perubahan
1.0.0	01 Januari 2026	-	Initial document
1.1.0	30 Januari 2026	-	Update API endpoints dan spesifikasi data sesuai implementasi backend (Sanic/Python)

Dokumen ini bersifat rahasia dan hanya untuk penggunaan internal.

Syntax error in text
mermaid version 10.9.5